Yifei Zhang

Applied Scientist, Amazon Alexa AI | Boston, Massachusetts
jimmyzyf@amazon.com · yifei.zhang2@emory.edu
yifeizhangcs.github.io · Google Scholar · GitHub

Last updated: November 14, 2026

Education

Emory University Aug. 2022 – Mar. 2026
- Ph.D. in Computer Science
- Advisor: Prof. Liang Zhao
- Dissertation: Benchmarking, Correcting, and Accelerating AI Reasoning across Diverse Modalities

Columbia University Sept. 2020 – Dec. 2021
- M.S. in Data Science

Duke University Aug. 2019 – May 2020
- Master of Engineering Management

Dalian University of Technology Sept. 2015 – June 2019
- B.E. in Engineering Mechanics

Research Highlights

- **AI Agents and Evaluation.** Develop benchmarks and evaluation methods for long-horizon dialogue, goal tracking, memory, and multimodal assistant behavior, including ATOD and Hear2Act.
- **Explainable and Trustworthy AI.** Build explanation-guided learning methods and benchmarks for evaluating visual explanations, including MAGI, ELAD, and Saliency-Bench.
- **Multimodal Reasoning.** Study retrieval-augmented image generation, cognitive-emotional reasoning, and agentic medical reasoning across language, vision, and clinical data.

Research and Industry Experience

Amazon Apr. 2026 – Present
Applied Scientist Boston, MA
- Develop scalable evaluation methods for agentic conversational systems, covering reliability, task completion, conversational quality, and multimodal behavior.
- Combine diverse evaluation signals into interpretable quality assessments for model and system improvement.
- Research privacy-aware model adaptation and distillation under practical system constraints.

Amazon May 2025 – Aug. 2025
Applied Scientist Intern Boston, MA
- Developed a memory system and multi-agent evaluation framework for agentic conversational systems.
- Built synthetic dialogue-generation and benchmarking pipelines for long-horizon, multi-turn, and concurrent user tasks.

Amazon AGI June 2024 – Sept. 2024
Research Scientist Intern Boston, MA
- Developed reward modeling and preference-based evaluation workflows for LLM-driven conversational interactions.
- Built synthetic preference-data generation and parameter-efficient fine-tuning pipelines.

Emory University Aug. 2022 – Mar. 2026
Ph.D. Student Researcher Atlanta, GA
- Developed evaluation benchmarks and explanation-guided methods for reliable AI across language, vision, and multimodal data.

Publications

Full publication list: Google Scholar.

Peer-Reviewed and Accepted Papers

Dual Decomposition for Text-to-Image Retrieval and Generation EACL Findings, 2027 (accepted)

M. Zhu, S. Cheng, G. Bai, **Yifei Zhang**, and L. Zhao

Beyond Efficiency: A Systematic Survey of Resource-Efficient Large Language Models ACM CSUR, 2026

G. Bai, Z. Chai, C. Ling, S. Wang, J. Lu, N. Zhang, T. Shi, Z. Yu, M. Zhu, **Yifei Zhang**, X. Song, C. Yang, Y. Cheng, and L. Zhao

ATOD: An Evaluation Framework and Benchmark for Agentic Task-Oriented Dialogue Systems AACL-IJCNLP, 2026

Yifei Zhang, H. Nayyeri, R. Khaziev, E. Yilmaz, G. Tur, D. Hakkani-Tür, and H. P. Thadakamalla

Hear2Act: Benchmarking When Prosody Should Change What an Assistant Does NeurIPS RTCA, 2026 (workshop poster)

X. Liu, H. Nayyeri, D. Hakkani-Tür, E. Yilmaz, J.-K. Kim, **Yifei Zhang**, C. Peris, and H. Thadakamalla

Saliency-Bench: A Comprehensive Benchmark for Evaluating Visual Explanations KDD, 2025

Yifei Zhang, J. Song, S. Gu, T. Jiang, B. Pan, G. Bai, and L. Zhao

GraphNarrator: Generating Textual Explanations for Graph Neural Networks ACL, 2025

B. Pan, Z. Xiong, G. Wu, Z. Zhang, Y. Hu, **Yifei Zhang**, and L. Zhao

Distilling Large Language Models for Text-Attributed Graph Learning CIKM, 2024

B. Pan, Z. Zhang, **Yifei Zhang**, Y. Hu, and L. Zhao

Multi-Prompt Fine-Tuning of Foundation Models for Enhanced Biomedical Image Segmentation Springer, 2024 (book chapter)

X. Li, **Yifei Zhang**, and L. Zhao

DUE: Dynamic Uncertainty-Aware Explanation Supervision via 3D Imputation KDD, 2024

Q. Zhao, **Yifei Zhang**, M. Zhu, S. Gu, Y. Gao, X. Yang, and L. Zhao

ELAD: Explanation-Guided Large Language Models Active Distillation ACL Findings, 2024

Yifei Zhang, B. Pan, C. Ling, Y. Hu, and L. Zhao

Visual Attention Prompted Prediction and Learning IJCAI, 2024

Yifei Zhang, B. Pan, S. Gu, G. Bai, M. Qiu, X. Yang, and L. Zhao

STES: A Spatiotemporal Explanation Supervision Framework SDM, 2024

D. Yu, B. Chen, Y. Li, S. Dhakal, **Yifei Zhang**, Z. Liu, M. Zhang, J. Zhang, and L. Zhao

MAGI: Multi-Annotated Explanation-Guided Learning ICCV, 2023

Yifei Zhang, S. Gu, Y. Gao, B. Pan, X. Yang, and L. Zhao

ESSA: Explanation Iterative Supervision via Saliency-Guided Data Augmentation KDD, 2023

S. Gu, **Yifei Zhang**, Y. Gao, X. Yang, and L. Zhao

Preprints

Agentic Commerce: A Survey of How AI Agents Are Reshaping Commerce TechRxiv, 2026
Yifei Zhang, B. Pan, M. Zhu, J. Pei, and L. Zhao

EmoLLM: Appraisal-Grounded Cognitive-Emotional Co-Reasoning in Large Language Models arXiv, 2026
Yifei Zhang, M. Li, H. Gao, and L. Zhao

Explainable Cross-Disease Reasoning for Cardiovascular Risk Assessment from Low-Dose Computed Tomography arXiv, 2025
Yifei Zhang, J. Zhang, M. Safari, X. Yang, and L. Zhao

MEGL: Multimodal Explanation-Guided Learning arXiv, 2024
Yifei Zhang, T. Jiang, B. Pan, J. Wang, G. Bai, and L. Zhao

Saliency-Guided Hidden Associative Replay for Continual Learning arXiv, 2023
G. Bai, Q. Zhao, X. Jiang, **Yifei Zhang**, and L. Zhao

SurroCBM: Concept Bottleneck Surrogate Models for Generative Post-hoc Explanation arXiv, 2023
B. Pan, Z. Liu, **Yifei Zhang**, and L. Zhao

Controllable Data Generation Via Iterative Data-Property Mutual Mappings arXiv, 2023
B. Pan, M. Qin, S. Wang, **Yifei Zhang**, and L. Zhao

Academic Service

Journal Reviewer: ACM TIST; IEEE TBD; IEEE TNNLS; JACMP; Expert Systems with Applications; Information Processing and Management; Neurocomputing; Information Systems Journal.
Conference Reviewer / Program Committee: ICLR 2025–2027; NeurIPS 2026; AAAI 2027; CVPR 2025–2026; ICCV 2025; ECCV 2026; BMVC 2026; ACL ARR 2025–2026; KDD Research Track 2025–2026; KDD Datasets and Benchmarks Track 2025–2026; MemAgents Workshop (ICLR 2026).
Workshop Organization: Publicity Chair, DeepSpatial 2024 (KDD); Student Co-organizer, FM4sBio 2025 (AAAI).

Honors, Grants, and Memberships

- Sigma Xi, Full Membership.
- Tinker Research Grant.
- KDD 2023 Student Travel Award.
- Dalian University of Technology Overseas Alumni Association Seagle Scholarship Award.
- Academic Excellence Scholarship Award, Dalian University of Technology.
- Technological Innovation Scholarship Award, Dalian University of Technology.

Technical Skills

Programming: Python, C/C++, MATLAB, Java, Shell
AI / LLM Frameworks: PyTorch, Transformers, TensorFlow, vLLM, verl, TRL, PEFT
Training and Infrastructure: Accelerate, DeepSpeed, AWS, Docker, Linux, Git